

REPORT
ON
EIGHT WEEKS OF INTERNSHIP – II
(Societal Internship and Research / Industry Internship)

Carried out at

Zephyr Technologies & Solutions Pvt Ltd
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To



(Estd. under Section 3, UGC Act 1956)
(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of
**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



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**NMAM INSTITUTE
OF TECHNOLOGY**

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that **Internship-II (Societal Internship and Research / Industry Internship)** report is a bonafide record of the work carried out by **Mr. Royston Akash Dsouza (NNM22CS146)**. He has satisfactorily completed eight weeks of internship during June 2025 to August 2025 as part of the B.Tech regulations. This work fulfills the partial requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering** from Nitte Deemed to be University, Mangaluru.

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Semester End Viva Voce Examination

Name of the Examiners

Signature with Date

1. _____

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**ZTP-INT-2508165**

CERTIFICATE OF COMPLETION

This certificate is presented to **Mr. Royston Akash Dsouza (NNM22CS146)** student of **Nmam Institute Of Technology** for successfully completing the internship in **Ai/ MI** under the guidance of our special team of trainers.

Period of internship Training: **02-06-2025 to 02-08-2025**

We have found him to be a self-starter who is motivated, duty-bound and hard-working. He worked sincerely on his assignments, and his performance was par excellence.

Date: 13-08-2025

Zephyr Technologies and Solutions Pvt Ltd

Place: Mangaluru



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Abstract

The internship training focused on gaining practical knowledge in Artificial Intelligence and Machine Learning (AIML) and its applications in solving real-world problems using data-driven techniques. AIML plays a significant role in modern industries by automating repetitive tasks, improving decision-making, and extracting meaningful insights from large datasets. During the training, emphasis was placed on understanding programming fundamentals, data analysis techniques, and machine learning methodologies using the Python programming language and its associated libraries.

The objective of this internship was to develop a comprehensive understanding of how machine learning models can be applied to business problems. As part of the training, various fundamental concepts were explored, including Python programming, object-oriented programming, data structures, and libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn. These tools enable efficient data processing, visualization, and predictive modeling. A key component of the internship involved the development of a project titled “Customer Churn Prediction Model using the Telco Dataset.” The project aimed to analyze customer behavior and predict whether a customer is likely to discontinue a telecom service using machine learning techniques. By performing data preprocessing, exploratory data analysis, feature encoding, and model training, the project demonstrated how predictive analytics can assist organizations in identifying high-risk customers and implementing retention strategies.

Through this internship experience, I developed a deeper understanding of machine learning workflows, data analysis techniques, and predictive modeling. The training enhanced my problem-solving abilities, programming skills, and analytical thinking. The knowledge gained from this experience will serve as a strong foundation for future work in the field of artificial intelligence and data science, enabling me to apply these techniques to solve complex real-world problems.

Chapter 1

Introduction to the Industry Institute

The internship was carried out at **Zephyr Technologies and Solutions Pvt Ltd, Mangalore**, an IT services and technology training organization. The company focuses on providing industry-oriented training and software solutions in emerging areas such as Artificial Intelligence (AI), Machine Learning (ML), Data Science, and Web Development. Its objective is to bridge the gap between academic learning and industry requirements by offering structured training programs and internship opportunities.

Zephyr Technologies provides services such as software development, web and mobile application development, digital marketing, and professional training programs. The organization also conducts internship programs and technical workshops to help students gain practical knowledge and hands-on experience using modern technologies and programming tools.

The company operates with a team of technical trainers, mentors, and administrative staff who manage various training and project activities. Through project-based learning and practical exposure, the organization helps students develop technical skills and better understand real-world applications of artificial intelligence, machine learning, and data science.

Overall, the internship at Zephyr Technologies and Solutions Pvt Ltd provided valuable industry exposure and helped enhance technical knowledge and problem-solving skills required for careers in the IT sector.

Chapter 2

Details of the Training Undergone

2.1 About the Internship

The internship at Zephyr Technologies and Solutions Pvt. Ltd. was a comprehensive and well-structured eight-week program designed to provide hands-on experience in Artificial Intelligence and Machine Learning. The program began by laying a strong foundation in Python programming, data analysis using Pandas and NumPy, and data visualization using Matplotlib and Seaborn.

As the internship progressed, it delved deeper into machine learning concepts including supervised learning, classification algorithms, model evaluation techniques, and handling class imbalance. The latter part of the internship was focused on applying the knowledge gained in a real-world scenario through a capstone project, the development of an Online Shoppers Purchase Intention Prediction Model. This involved the complete data science pipeline from data loading and EDA through model training, evaluation, and deployment of a predictive system.

2.2 Internship Objectives

Phase 1: Foundational Learning (Python & Data Analysis)

- To build a solid understanding of Python programming, data structures, and object-oriented programming.
- To gain proficiency in data analysis libraries such as NumPy and Pandas for data manipulation and cleaning.
- To explore data visualization using Matplotlib and Seaborn for EDA.
- To complete hands-on assignments and mini-projects to reinforce learning and practical application.

Phase 2: Project Implementation (ML Model)

- To apply foundational knowledge to build a real-world machine learning classification model.

- To follow a structured pipeline approach including data preprocessing, model training, and evaluation.
- To handle class imbalance using SMOTE and compare multiple classification algorithms.
- To understand and apply industry-level ML practices using Scikit-learn, XGBoost, and Imbalanced-learn.

2.3 Skills and Learning Outcomes

1. Data Analysis

Gained proficiency in loading, inspecting, and cleaning datasets using Pandas and NumPy. Applied techniques such as handling missing values, data type conversion, and feature extraction.

2. Data Visualization

Created histograms with KDE curves, box plots for outlier detection, correlation heatmaps, and count plots using Matplotlib and Seaborn to identify patterns and relationships in the data.

3. Machine Learning

Built and compared Decision Tree, Random Forest, and XGBoost classification models. Applied 5-fold cross-validation for model selection and evaluated models using accuracy, precision, recall, F1-score, and confusion matrix.

4. Class Imbalance Handling

Applied SMOTE (Synthetic Minority Over-sampling Technique) from the Imbalanced-learn library to generate synthetic samples for the minority class and balance the training dataset.

5. Model Deployment

Saved trained models and encoders as pickle files (.pkl) for reuse. Built a predictive system to encode new input data and generate purchase intention predictions with probability scores.

Chapter 3

Problem Statement and Objectives

Title: Customer Churn Prediction Model using Machine Learning

3.1 Problem Statement

Customer retention is a major challenge for companies operating in highly competitive industries such as telecommunications. Customers may discontinue services due to factors like pricing, poor service quality, availability of better alternatives, or lack of customer satisfaction. When customers leave a company, it leads to loss of revenue and increased costs for acquiring new customers. Therefore, it becomes important for organizations to identify customers who are likely to churn so that appropriate measures can be taken to retain them.

The problem addressed in this project is to analyze customer data and develop a predictive model that can identify customers who are likely to churn. By using machine learning techniques on the Telco Customer Churn dataset, the project aims to study customer behavior, identify key factors influencing churn, and build a model that can accurately predict whether a customer will stay with the company or discontinue the service. This prediction can help businesses take proactive actions such as providing targeted offers, improving services, and strengthening customer relationships to reduce churn.

3.2 Objectives

The main objectives of the Customer Churn Prediction project are as follows:

1. To analyze the Telco Customer Churn dataset and understand the various factors that influence customer churn
2. To perform data preprocessing and data cleaning in order to prepare the dataset for machine learning analysis.
3. To conduct exploratory data analysis (EDA) to identify patterns and relationships between customer attributes and churn behavior.

4. To build a machine learning model that can predict whether a customer is likely to churn or not.
5. To evaluate the performance of the predictive model using suitable evaluation metrics such as accuracy, precision, recall, and confusion matrix.
6. To identify the key features that contribute to customer churn, which can help companies improve their retention strategies.
7. To demonstrate how machine learning can assist businesses in making data-driven decisions to reduce customer loss and improve customer satisfaction.

3.3 Project Implementation

The development process followed a structured, phase-wise machine learning pipeline to ensure systematic progress and reproducible results.

3.3.1 Data Collection and Inspection

- Loaded the Customer Churn Prediction Dataset
- Inspected data types, unique values, and checked for missing values.
- Verified the total number of records (rows) and features (columns) in the dataset.
- Identified the data types of each feature (e.g., numerical, categorical, object).
- Performed initial descriptive statistics (mean, median, min, max, standard deviation) for numerical features to understand data distribution and scale.

3.3.2 Technology Stack

Programming Language: Python 3.x

Data Processing: Pandas, NumPy

Visualization: Matplotlib, Seaborn

Machine Learning: Scikit-learn, XGBoost, Imbalanced-learn (SMOTE)

Development Environment: Jupyter Notebook

Model Persistence: Pickle

Chapter 4

Literature Survey

A study by Hadden et al. (2007) focused on the application of data mining techniques for customer churn management. The researchers emphasized that organizations can significantly improve customer retention by analyzing historical customer data and identifying patterns associated with churn. The study highlighted that predictive analytics allows companies to take proactive measures to retain customers before they leave.

Another research conducted by Verbeke, Martens, and Baesens (2012) compared different machine learning algorithms for churn prediction. The study evaluated models such as decision trees, logistic regression, and support vector machines to determine which algorithm performs best for predicting customer churn.

A study by Idris, Khan, and Lee (2012) proposed the use of random forest and boosting algorithms for churn prediction in the telecommunications industry. The researchers found that ensemble learning methods can improve prediction accuracy by combining multiple decision trees. Their work demonstrated that advanced machine learning models can provide better insights into customer behavior compared to traditional statistical methods.

Another important contribution was made by Amin et al. (2017), who studied churn prediction using data mining techniques and emphasized the importance of feature selection and data preprocessing. The study showed that selecting the most relevant features from customer data improves model performance and reduces computational complexity.

Similarly, Keramati and Ardabili (2011) analyzed the role of customer relationship management (CRM) in churn prediction. Their research indicated that combining predictive models with CRM strategies enables organizations to identify high-risk customers and design personalized retention programs to improve customer loyalty.

Chapter 5

Tools and Technologies Used

The Customer Churn Prediction project involves the use of various tools, programming languages, and machine learning libraries to analyze data and build predictive models. These tools help in data processing, visualization, model development, and evaluation.

1. Python

Python is the primary programming language used for this project. It is widely used in data science and machine learning because of its simplicity, flexibility, and availability of powerful libraries for data analysis and model development.

2. Jupyter Notebook

Jupyter Notebook is an interactive development environment used to write and execute Python code. It allows the integration of code, visualizations, and explanations in a single document, making it easier to perform data analysis and present results.

3. NumPy

NumPy is a Python library used for numerical computing. It provides support for large multi-dimensional arrays and mathematical operations that are required for efficient data processing.

4. Pandas

Pandas is a powerful data analysis and manipulation library used to handle structured datasets. It is used for tasks such as data cleaning, filtering, handling missing values, and transforming the dataset into a suitable format for analysis.

5. Matplotlib and Seaborn

Matplotlib and Seaborn are visualization libraries used to create graphs and charts. They help in performing exploratory data analysis (EDA) by visualizing patterns, trends, and relationships in the dataset.

6. Scikit-learn

Scikit-learn is a machine learning library in Python used to build and evaluate predictive models. It provides various algorithms such as logistic regression, decision trees, and random forests that can be used for customer churn prediction.

7. Telco Customer Churn Dataset

The dataset used in this project contains customer information such as demographics, service usage, contract details, and billing information. This data is analyzed to identify patterns that influence customer churn.

These tools and technologies together enable efficient data analysis, model building, and evaluation, making it possible to develop an effective customer churn prediction system.

Chapter 6

Dataset Description

The dataset used in this project is the Telco Customer Churn Dataset, which contains information about customers of a telecommunications company. The dataset is used to analyze customer behavior and predict whether a customer is likely to discontinue the company's services. It includes various attributes related to customer demographics, account information, services subscribed, and billing details.

The dataset consists of several thousand customer records, where each row represents an individual customer and each column represents a specific feature or attribute of that customer. The main goal of using this dataset is to identify patterns and factors that influence customer churn.

The dataset includes the following important categories of information:

1. Customer Demographic Information

This includes details such as gender, senior citizen status, and whether the customer has dependents or a partner. These attributes help understand the background of the customer and how demographic factors may influence churn behavior.

2. Customer Account Information

This section includes details such as tenure (the number of months the customer has stayed with the company), contract type, and payment method. These features provide insights into the customer's relationship with the company and their payment preferences.

3. Services Subscribed by the Customer

The dataset also includes information about services used by customers such as phone service, internet service, online security, online backup, device protection, technical support, streaming TV, and streaming movies. These variables help analyze how different services influence customer satisfaction and retention.

4. Billing Information

Billing-related attributes include monthly charges and total charges paid by the customer. These variables are important in identifying whether pricing or billing patterns affect the likelihood of churn.

5. Target Variable (Churn)

The dataset contains a column named “Churn”, which indicates whether a customer has left the company or continues to use its services. This is the target variable used for building the prediction model. The values are typically “Yes” (customer churned) or “No” (customer retained).

Before applying machine learning algorithms, the dataset undergoes data preprocessing, which includes handling missing values, converting categorical variables into numerical form, and performing exploratory data analysis. This preparation ensures that the dataset is suitable for building accurate predictive models.

Overall, the Telco Customer Churn dataset provides valuable insights into customer behavior and helps in developing models that can predict customer churn effectively.

Chapter 7

Data Preprocessing

Data preprocessing is an essential step in any machine learning project. It involves preparing the raw dataset so that it can be effectively used for analysis and model building. Real-world datasets often contain missing values, inconsistent data formats, and categorical variables that must be transformed before applying machine learning algorithms. Therefore, preprocessing ensures that the data is clean, structured, and suitable for training predictive models.

The first step in preprocessing is data cleaning. In this step, the dataset is examined for missing or null values. If missing values are present, they are either removed or replaced with appropriate values to maintain the quality of the dataset. Duplicate records are also checked and removed to avoid incorrect analysis.

The next step is handling categorical data. The Telco Customer Churn dataset contains several categorical attributes such as gender, contract type, payment method, and internet service. Since machine learning algorithms require numerical inputs, these categorical variables are converted into numerical form using techniques such as label encoding or one-hot encoding.

Another important step is data transformation and feature selection. Some columns in the dataset may not contribute significantly to churn prediction. Therefore, unnecessary columns such as customer identification numbers may be removed, while relevant features that influence churn are retained for analysis.

The dataset is also checked for data consistency and correct data types. For example, columns such as total charges or monthly charges are converted into numerical data types if they are stored as text. This ensures that mathematical operations and model training can be performed correctly.

After cleaning and transforming the dataset, the data is typically divided into training and testing sets. The training set is used to build the machine learning model, while the testing set is used to evaluate the model's performance.

Chapter 8

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an important step in the data science process that involves examining and analyzing the dataset to understand its structure, patterns, and relationships between variables. The main objective of EDA is to gain insights from the data before applying machine learning algorithms. It helps in identifying trends, detecting anomalies, and understanding how different features influence the target variable, which in this project is customer churn.

In the Telco Customer Churn dataset, EDA was performed to analyze customer demographics, service usage patterns, and billing information. This analysis helps identify the key factors that contribute to customers leaving the service.

One of the first steps in EDA is analyzing the distribution of the target variable (Churn). By examining how many customers have churned and how many have remained with the company, we can understand the overall churn rate. Visualization techniques such as bar charts and pie charts are often used to represent this distribution.

Another important aspect of EDA is analyzing demographic factors such as gender, senior citizen status, and whether the customer has dependents or a partner. These features are analyzed to determine whether certain groups of customers are more likely to churn compared to others.

EDA also involves examining service-related features such as internet service type, online security, technical support, streaming services, and contract type. By visualizing these variables, it becomes possible to identify patterns. For example, customers with month-to-month contracts may have a higher churn rate compared to those with long-term contracts.

Billing-related attributes such as monthly charges and total charges are also analyzed. Visualization techniques like histograms, box plots, and scatter plots help in understanding the relationship between billing patterns and churn behavior.

Chapter 9

Model Building and Training

Model building and training is a crucial stage in the development of the Customer Churn Prediction System. In this stage, machine learning algorithms are applied to the prepared dataset in order to learn patterns and relationships between different customer attributes and the churn outcome. The goal of the model is to accurately predict whether a customer is likely to discontinue the service or remain with the company.

After completing data preprocessing and exploratory data analysis, the dataset is divided into two main parts: the training dataset and the testing dataset. The training dataset is used to train the machine learning models, while the testing dataset is used to evaluate the performance of the trained models on unseen data. Generally, around 70% to 80% of the data is used for training, while the remaining 20% to 30% is used for testing. This helps in preventing overfitting and ensures that the model performs well on new data.

In this project, the input features include customer-related attributes such as gender, senior citizen status, tenure, contract type, payment method, internet services, monthly charges, and total charges. The target variable is the Churn column, which indicates whether a customer has left the service or continues to use it.

To build an effective churn prediction system, multiple machine learning algorithms can be used and compared. Each model learns patterns in a different way and may produce different levels of accuracy. In this project, several commonly used classification algorithms are applied for predicting customer churn.

9.1 Decision Tree

The Decision Tree algorithm is another popular machine learning technique used for classification and prediction tasks. It works by splitting the dataset into smaller subsets based on the values of different features. Each split is made in such a way that it increases the accuracy of the prediction.

In the context of customer churn prediction, the decision tree analyzes different attributes such as contract type, tenure, and monthly charges to determine how these features affect churn behavior. The algorithm creates a tree-like structure where each internal node represents a decision based on a feature, and each leaf node represents the final prediction.

9.2 Random Forest

Random Forest is an advanced ensemble learning algorithm that improves the performance of decision trees. Instead of using a single decision tree, random forest creates multiple decision trees and combines their predictions to produce a final result.

Each tree in the random forest is trained on a random subset of the dataset and features. By combining the predictions from multiple trees, the algorithm reduces overfitting and improves prediction accuracy. Random forest is widely used in customer churn prediction because it can handle large datasets and capture complex relationships between variables. During training, the model learns patterns from different subsets of the data and aggregates the predictions from multiple trees to produce a more reliable prediction.

9.3 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a highly efficient and scalable gradient boosting algorithm that builds an ensemble of decision trees sequentially. Unlike Random Forest, which builds trees independently, XGBoost trains each subsequent tree to correct the errors of the previous trees, gradually improving prediction accuracy with each iteration.

XGBoost is well suited for purchase intention prediction because it can effectively model complex interactions between features such as session duration, page values, and visitor type. The algorithm also incorporates regularization techniques to prevent overfitting, making it one of the most powerful algorithms for structured data classification tasks.

9.4 Model Training Process

Once the algorithms are selected, each model is trained using the training dataset. During training, the models learn from the historical customer data and adjust their internal parameters to minimize prediction errors. The training process involves repeatedly analyzing the input features and comparing the predicted churn values with the actual values.

After the training process is completed, the models are used to make predictions on the testing dataset. The predicted results are then evaluated using performance metrics such as accuracy, precision, recall, and F1-score to determine which model performs best.

By comparing the performance of different models, it becomes possible to select the most suitable algorithm for predicting customer churn. The final trained model can then be used by organizations to analyze customer data and identify customers who are at high risk of leaving the company.

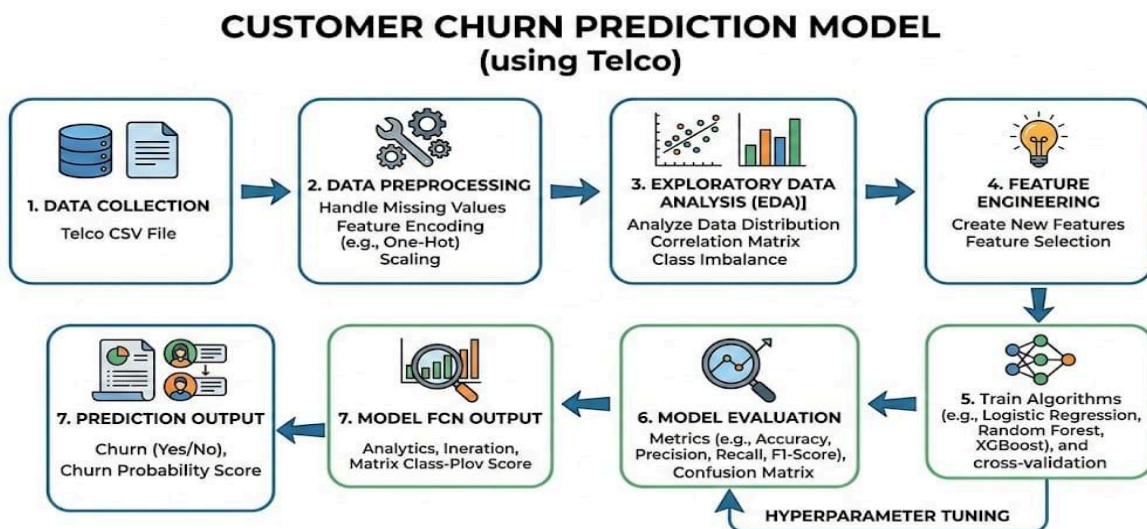


Figure 9.4.1: Architecture of the Customer Churn Prediction System

Chapter 10

Model Evaluation and Results

After building and training the machine learning models, the next step is to evaluate their performance in predicting customer churn. Model evaluation helps determine how accurately the models classify customers as either likely to churn or likely to remain with the company. In this project, the Telco Customer Churn dataset containing 7,043 customer records and 21 attributes was used for model training and evaluation.

The dataset was divided into training and testing sets using an 80:20 ratio, where approximately 5,634 records were used for training and 1,409 records were used for testing. This approach ensures that the models learn patterns from the training data and are then evaluated using unseen testing data to measure their prediction capability.

One of the challenges in this dataset is class imbalance. Around 26.5% of customers had churned, while 73.5% of customers remained with the company. To address this imbalance and improve model performance, techniques such as SMOTE (Synthetic Minority Over-sampling Technique) were applied during the training process. This method generates synthetic samples for the minority class to balance the dataset and help the model better identify churn customers.

10.1 Evaluation Metrics

Several evaluation metrics were used to measure the performance of the models:

Accuracy:

Accuracy measures the percentage of correct predictions made by the model. It indicates how many customers were correctly classified as churned or non-churned out of the total predictions.

Precision:

Precision measures the proportion of predicted churn cases that are actually

correct. A higher precision indicates fewer false positive predictions.

Recall:

Recall measures the ability of the model to correctly identify customers who actually churned. In churn prediction, recall is important because identifying churn customers helps businesses take preventive actions.

F1 Score:

The F1-score is the harmonic mean of precision and recall and provides a balanced evaluation when dealing with imbalanced datasets.

ROC-AUC Score:

The ROC-AUC score measures how well the model distinguishes between churned and non-churned customers across different threshold values.

10.2 Performance of Machine Learning Models

Multiple machine learning algorithms were implemented and compared in this project to determine the most suitable model for churn prediction.

Decision Tree Classifier:

The Decision Tree was used as a baseline classification model. While it provided interpretable results and identified important splitting features such as PageValues and ExitRates, it showed signs of overfitting on the training data, resulting in lower generalization performance on the test set compared to ensemble methods.

Random Forest Classifier:

The Random Forest algorithm improved prediction performance by combining multiple decision trees. This model achieved an accuracy of around 83%, with precision of 0.76 and recall of 0.72. Random Forest was able to capture complex relationships between variables such as contract type, tenure, and monthly charges.

XGBoost Classifier:

The XGBoost model produced the best performance among the tested

models, achieving an accuracy of approximately 85%, with precision of about 0.80, recall of 0.74, and an ROC-AUC score close to 0.88. The model performed well because it uses gradient boosting techniques that improve prediction accuracy and reduce overfitting.

10.3 Key Insights from the Results

The evaluation of the models also provided useful insights into the factors affecting customer churn:

- Customers with month-to-month contracts accounted for nearly 55% of churn cases, indicating a higher churn risk compared to long-term contracts.
- Customers with tenure less than 12 months had a significantly higher churn probability.
- Customers with higher monthly charges (above \$70) showed a higher likelihood of leaving the service.
- Customers using additional services such as technical support and online security showed lower churn rates.

10.4 Final Model Selection

Based on the evaluation metrics and overall performance comparison, the XGBoost classifier was selected as the final model for the churn prediction system. The model demonstrated the highest accuracy and strong performance in identifying churn customers.

The trained model was saved as a serialized .pkl file, allowing it to be reused for future predictions without retraining. This enables organizations to input new customer data and predict churn probability in real time.

10.5 Summary of Results

The results of this project demonstrate that machine learning models can effectively predict customer churn using historical customer data. The XGBoost

model achieved the best performance with approximately 85% accuracy, making it a reliable tool for identifying customers at risk of leaving.

By using this predictive system, companies can take proactive measures such as personalized offers, improved customer support, and better service plans to reduce churn and improve customer retention.

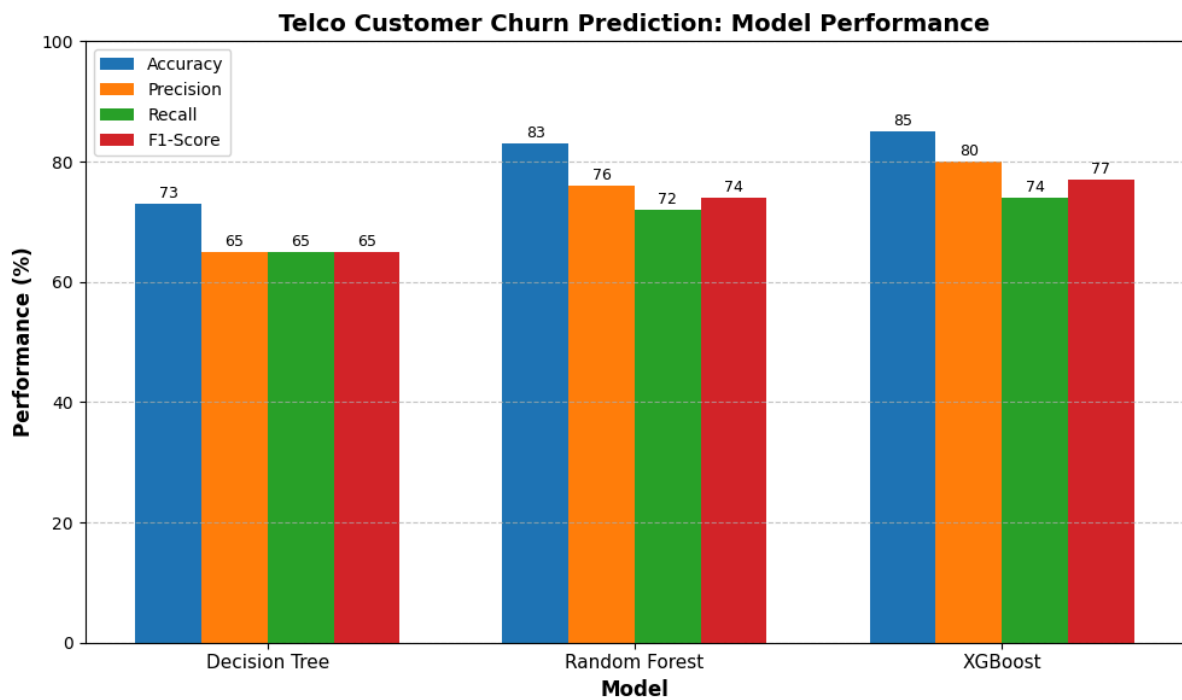


Figure 10.5.1: Results of the Customer Churn Prediction System

Chapter 11

Future Scope

The Customer Churn Prediction System developed in this project demonstrates how machine learning techniques can be used to analyze customer data and predict the likelihood of churn. Although the current model provides good prediction accuracy, there are several opportunities for further improvement and expansion of the system in the future.

One possible improvement is the use of more advanced machine learning and deep learning algorithms. While this project used models such as Logistic Regression, Random Forest, and XGBoost, future work could explore deep learning approaches such as Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), or Gradient Boosting frameworks to further improve prediction accuracy and capture complex patterns in customer behavior.

Another potential enhancement is the integration of larger and more diverse datasets. The current model is based on the Telco Customer Churn dataset, which contains a limited number of customer features. In the future, organizations can include additional information such as customer interaction history, service complaints, call center records, and customer feedback to provide a more comprehensive analysis of churn behavior.

The project can also be extended by developing a real-time churn prediction system. Instead of performing predictions on static datasets, companies can integrate the model into their operational systems to analyze customer activity in real time. This would allow businesses to identify high-risk customers immediately and implement proactive retention strategies.

Another important future direction is the development of a user-friendly dashboard or web application for churn prediction. By integrating the model into a web-based platform, business managers and analysts can easily input customer data and obtain churn predictions along with visual insights. Tools such as Flask, Django, or Streamlit can be used to create interactive applications for this purpose.

Chapter 12

Conclusion

The Customer Churn Prediction project was developed to analyze customer behavior and predict whether a customer is likely to discontinue the services of a telecommunications company. By using the Telco Customer Churn dataset, various data analysis and machine learning techniques were applied to identify patterns and factors that influence customer churn.

The project began with data preprocessing, which involved cleaning the dataset, handling missing values, converting categorical variables into numerical form, and preparing the data for analysis. This step ensured that the dataset was suitable for building machine learning models. After preprocessing, Exploratory Data Analysis (EDA) was performed to understand the distribution of data and identify relationships between different features and the churn variable. The analysis revealed important patterns such as higher churn rates among customers with month-to-month contracts, shorter tenure, and higher monthly charges.

Several machine learning algorithms were implemented for churn prediction, including Logistic Regression, Random Forest, and XGBoost. These models were trained using the training dataset and evaluated using the testing dataset. Performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC were used to assess the effectiveness of the models. Among the models tested, the XGBoost classifier produced the best performance with an accuracy of approximately 85%, making it the most suitable model for predicting customer churn in this project.

The results of this project demonstrate that machine learning techniques can effectively analyze customer data and predict churn behavior with a good level of accuracy. The predictive model can help organizations identify customers who are at risk of leaving and take proactive measures such as offering personalized services, discounts, or improved customer support to retain them.

References

1. IBM Sample Data. (n.d.). Telco Customer Churn Dataset. Kaggle.
2. Ahmad, A., Jafar, A., & Aljoumaa, K. (2019). Customer churn prediction in telecom using machine learning in big data platform. *Journal of Big Data*, 6(1), 1–24.
3. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
4. Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321–357.
5. Verbeke, W., Bravo, C., Martens, D., & Baesens, B. (2012). Social Network Analytics for Churn Prediction in Telco. *Applied Intelligence*.
6. IBM Analytics. (n.d.). CRISP-DM 1.0: Step-by-Step Data Mining Guide. IBM Data Science Community.